

TM Forum Technical Specification

Service Inventory

TMFC008

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Notice

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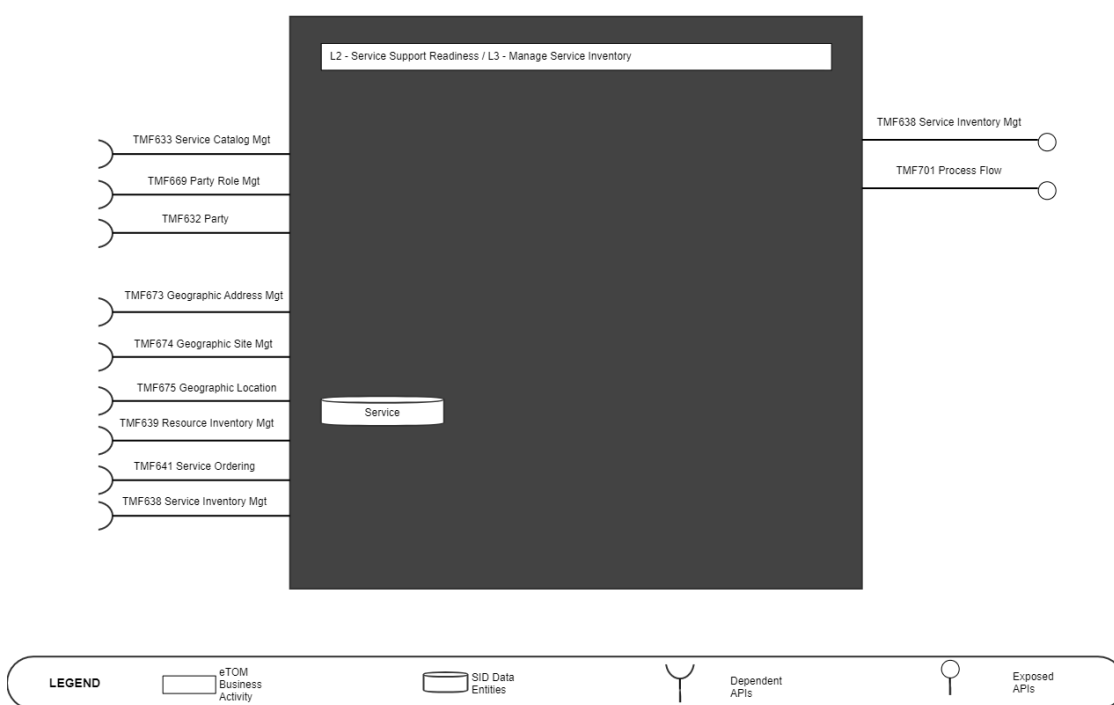
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1. Overview

Component Name	ID	Description	ODA Function Block
Service Inventory	TMFC008	Service Inventory component is responsible for storage and exposure of CFS (Customer Facing Service) that are associated to Product Inventory items. It is also responsible for RFS (Resource Facing Service) definition, mapping between CFS and RFS and mapping with infrastructure/network resources. Service Inventory component has functionality that enables inventory items creation, inventory organization, inventory search or filter, inventory monitoring and tracking, inventory control and inventory auditing. The minimum check to be performed at inventory items creation or update is the global consistency with related Service Catalog information.	Production

TMFC008 : Service Inventory



2. eTOM Processes SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

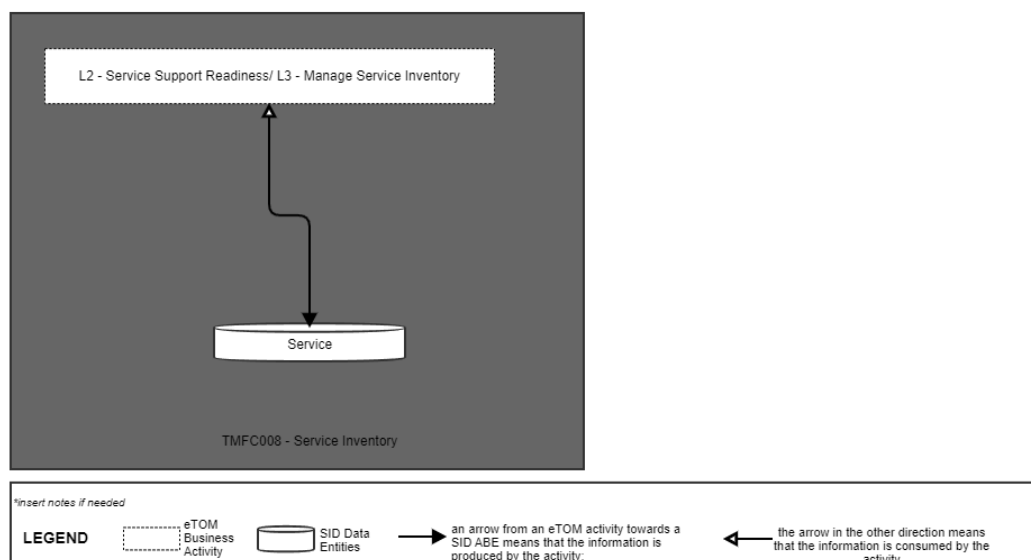
Identifier	Level	Business Activity Name	Description
1.4.4	L2	Service Support Readiness	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes
1.4.4.1	L3	Manage Service Inventory	Establish, manage and administer the enterprise's service inventory, as embodied in the Service Inventory Database, and monitor and report on the usage and access to the service inventory, and the quality of the data maintained in it.

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)
Service ABE	

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Functions Level 1	Aggregate Functions Level 2
576	Service Data Retrieval	Service Data Retrieval provides retrieval of appropriate inventory data for example in the context of service end to end testing.	Service Management	Service Repository Management
593	ServiceInventory Repository Updating	ServiceInventory Repository Updating updates information in the service inventory according to the configuration of specific services	Service Management	ServiceInventory Repository Management
628	Service to Resource Relationship Management	Service to Resource Relationship Management provides Creation, Update and Deletion of the relations of stand-alone physical or logical resources whose assignment is critical to service's fulfillment, and	Service Management	ServiceInventory Repository Management

Function ID	Function Name	Function Description	Aggregate Functions Level 1	Aggregate Functions Level 2
		whose tracking is critical to service operations, assurance, and billing, as well as, resources, which represent a larger resource structure supporting the service, often referred to as an Access Point.		
629	Service to Resource Relationship Synchronization	Service to Resource Relationship Synchronization function entails reconciliation of the data in a Service Inventory Management system with inventory discovered from other sources and synchronizes mismatched service inventory records.	Service Management	ServiceInventory Repository Management
630	Service-Resource Relationship Management Notifications	Service-Resource Relationship Management Notifications; Notification of Service-Resource Relationship Management actions to relevant stakeholders	Service Management	Service Reporting Service Repository Management
964	Onboarded Service Integration Configuration	Onboarded Service Integration Configuration function will configure the on boarded service and the relevant systems to establish integration automatically, when requested. There are several system services in the infrastructure that needs to be aware and integrated with the new service.	Service Management	Service Repository Management
965	Service Instance Lifecycle Management	Service Instance Lifecycle Management function will control the starting of new instances and closing of instances of a service as well as other activity states of the service instances. Software based Services' performance and availability may be controlled by managing multiple instances of the	Service Management	ServiceInventory Repository Management

Function ID	Function Name	Function Description	Aggregate Functions Level 1	Aggregate Functions Level 2
		service with multiple states of activity.		
1344	Service Topology Discovery	Service Topology Discovery function provides the required capability to discover how resources (e.g. network) are related to each other in providing a service.	Service Management	Service Repository Management

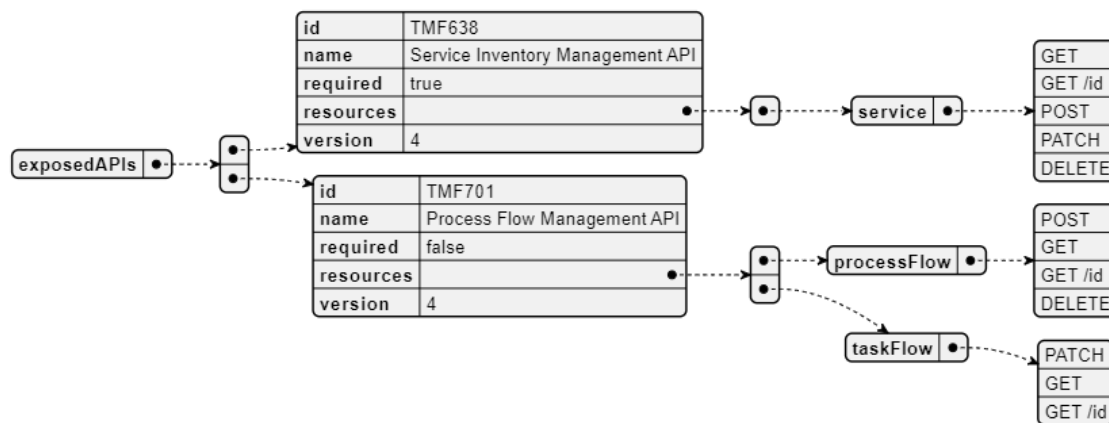
3. TMF OPEN APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

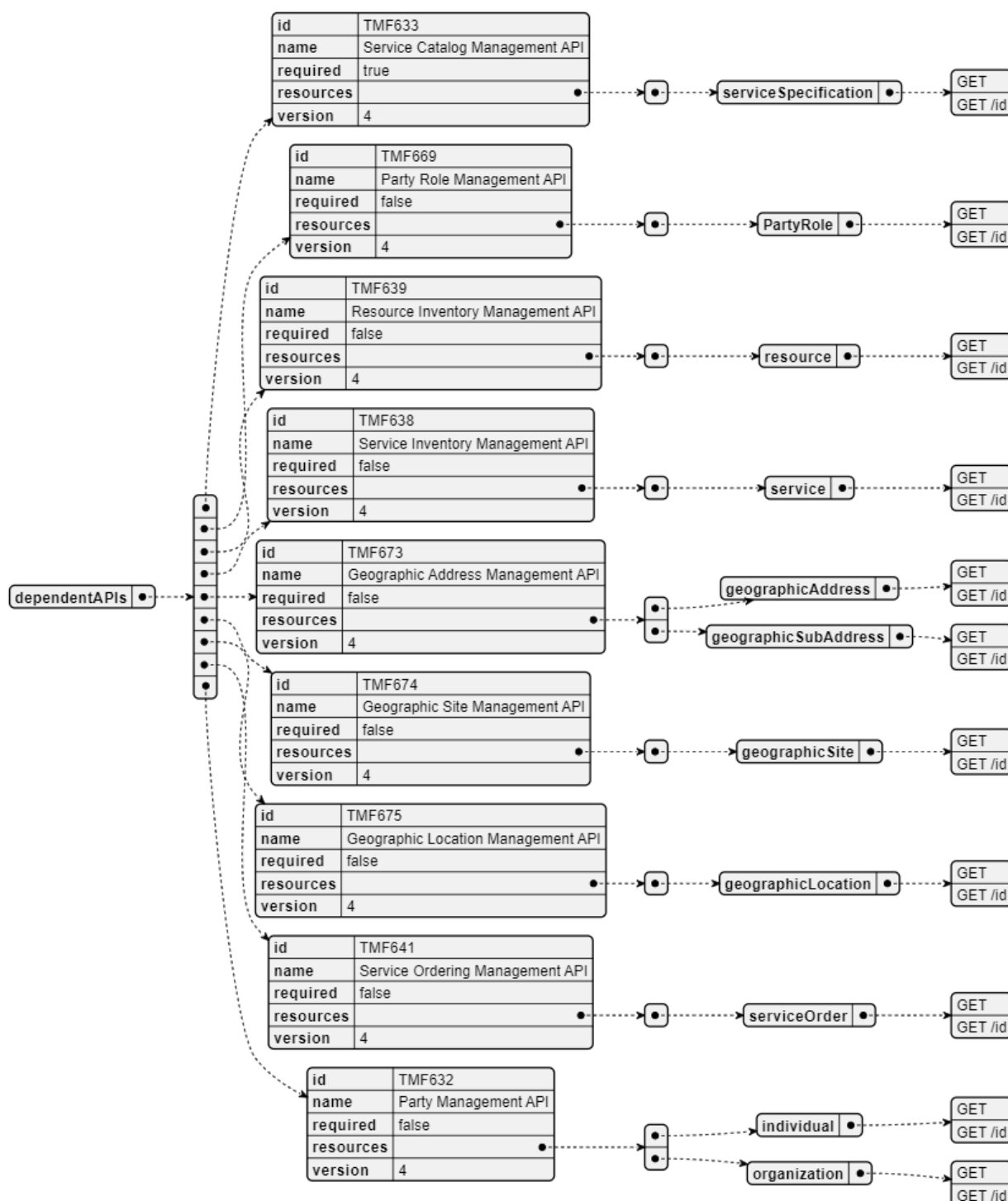
The following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF638	Service Inventory Management	4	Mandatory	service	Get Get /ID POST PATCH DELETE
TMF701	Process Flow	4	Optional	processFlow	Get Get /ID POST DELETE
				taskFlow	Get Get /ID PATCH
TMF688	Event	4	Optional	listener	POST
		4		hub	POST DELETE

3.2. Dependent APIs

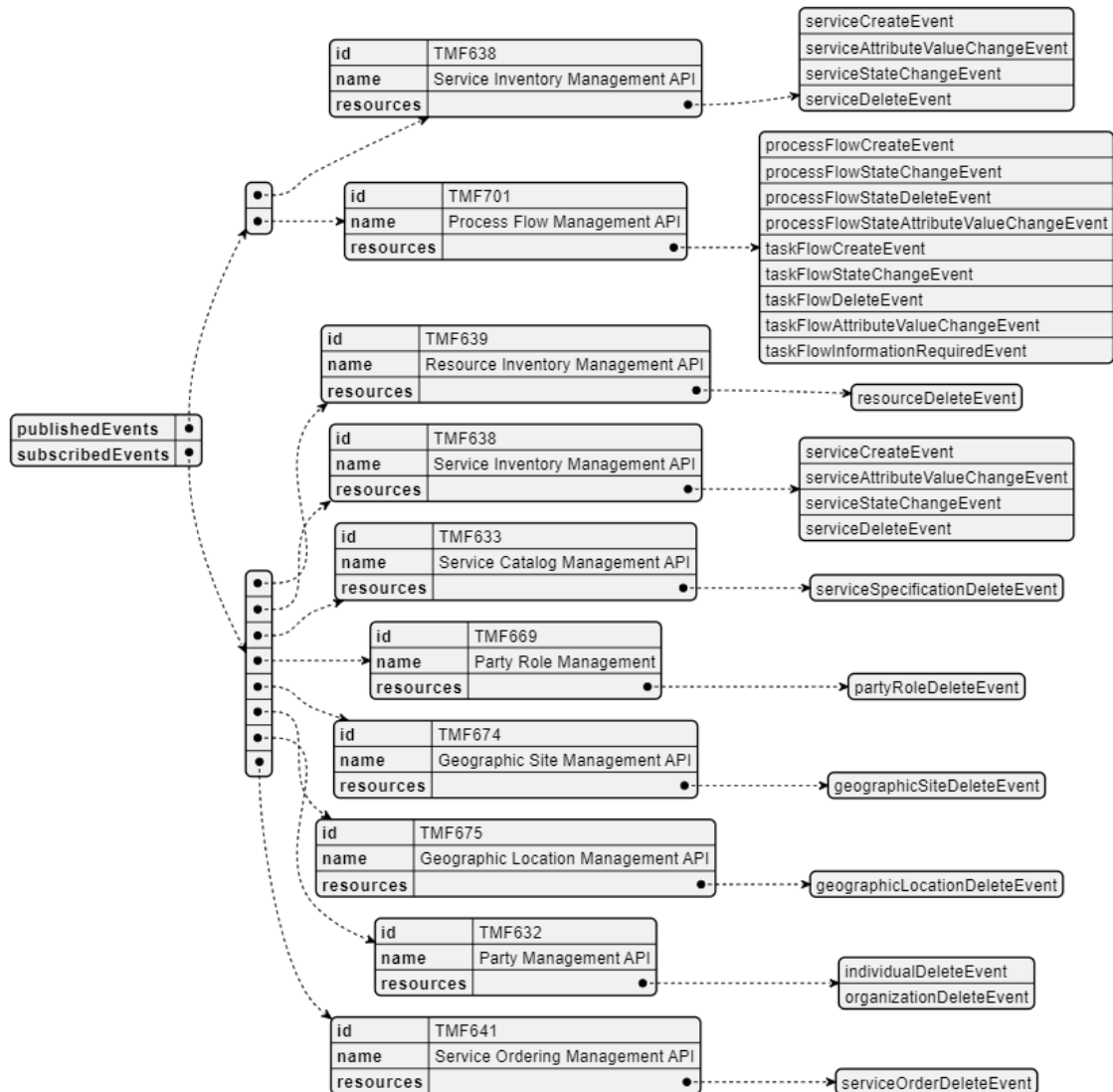
The following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Resources	Operations
TMF639	Resource Inventory Management	4	Optional	resource	Get Get /ID
TMF669	Party Role Management	4	Optional	partyRole	Get Get /ID
TMF632	Party Management	4	Optional	individual / organization	Get Get /ID
TMF672	User Roles Permission	4	Optional	permission	Get Get /ID
TMF673	Geographic Address Management	4	Optional	geographicAddress	Get Get /ID
				geographicSubAddress	Get Get /ID
TMF674	Geographic Site Management	4	Optional	geographicSite	Get Get /ID
TMF675	Geographic Location	4	Optional	geographicLocation	Get Get /ID
TMF633	Service Catalog Management	4	Mandatory	serviceSpecification	Get Get /ID
TMF641	Service Ordering	4	Optional	serviceOrder	Get Get /ID
TMF638	Service Inventory Management	4	Optional	service	Get Get /ID, Post, Patch, Delete
TMF688	Event	4	Optional	event	Get Get /ID, Post

3.3. Events

The following diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component Map on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	23.5
eTOM	23.5
Functional Framework	23.5

5.2. Jira References

5.2.1. Functional Framework

- [ISA-416](#) - Harmonize functions related to Product, Service and Resource Inventories **BACKLOG** Harmonize as much as possible functions related to Product, Service and Resource Inventory (this is under review)
- [ISA-996](#) - Master Data Management has repurposed catalog related functions **BACKLOG**

As part of reviewing the Product, Service and Resource Catalogs to align with the updates to the Functional Framework, we realized that a lot of functions have either been removed or been re-purposed by Master Data Management.

Due to this, we had to make a decision to remove all these functions from the Components:

- Removed deleted unclassified functions (1,2,5,7,9,10, 12,14,15)
- Removed functions that used to be related to Catalog management but now have been re-purposed by Master Data Management ((3,4, 6,11,13)

Our concern is that we may be missing functions for the catalog due to re-use of Master Data Management.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	31 May 2022	Kamal Maghsoudlou Sylvie Demarest	Initial release
1.0.1	27 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3.
1.0.1	15 Aug 2023	Amaia White	Final updates before publication
1.1.0	14 May 2024	Hugo Vaughan (TM Forum) Sylvie Demarest	- Updated to latest template - Aligned with 23.5 SID, eTOM and Functional Framework
1.1.0	12 Jul 2024	Amaia White	Final Updates before publication
1.2.0	12 Nov 2024	Gaetano Biancardi	- TMF688, removed from the core specification, moved to supporting functions - TMF672, removed from the core specification, moved to supporting functions - API version, only major version to be specified
1.2.0	26 Nov 2024	Amaia White	Final updates before publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	31 May 2022	Goutham Babu	Initial release
Production	22-Jul-2022	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	15-Aug-2023	Amaia White	new release v1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	26-Nov-2024	Amaia white	New release 1.2.0

Release Status	Date Modified	Modified by:	Description of changes
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.1.3. Acknowledgments

Team Member (@mention)	Company	Role*
Kamal Maghsoudlou	Ericsson	Editor
Sylvie Demarest	Orange	Additional Input
Ian Turkington	TM Forum	Additional Input
Hugo Vaughan (TM Forum)	TM Forum	Additional Input

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer